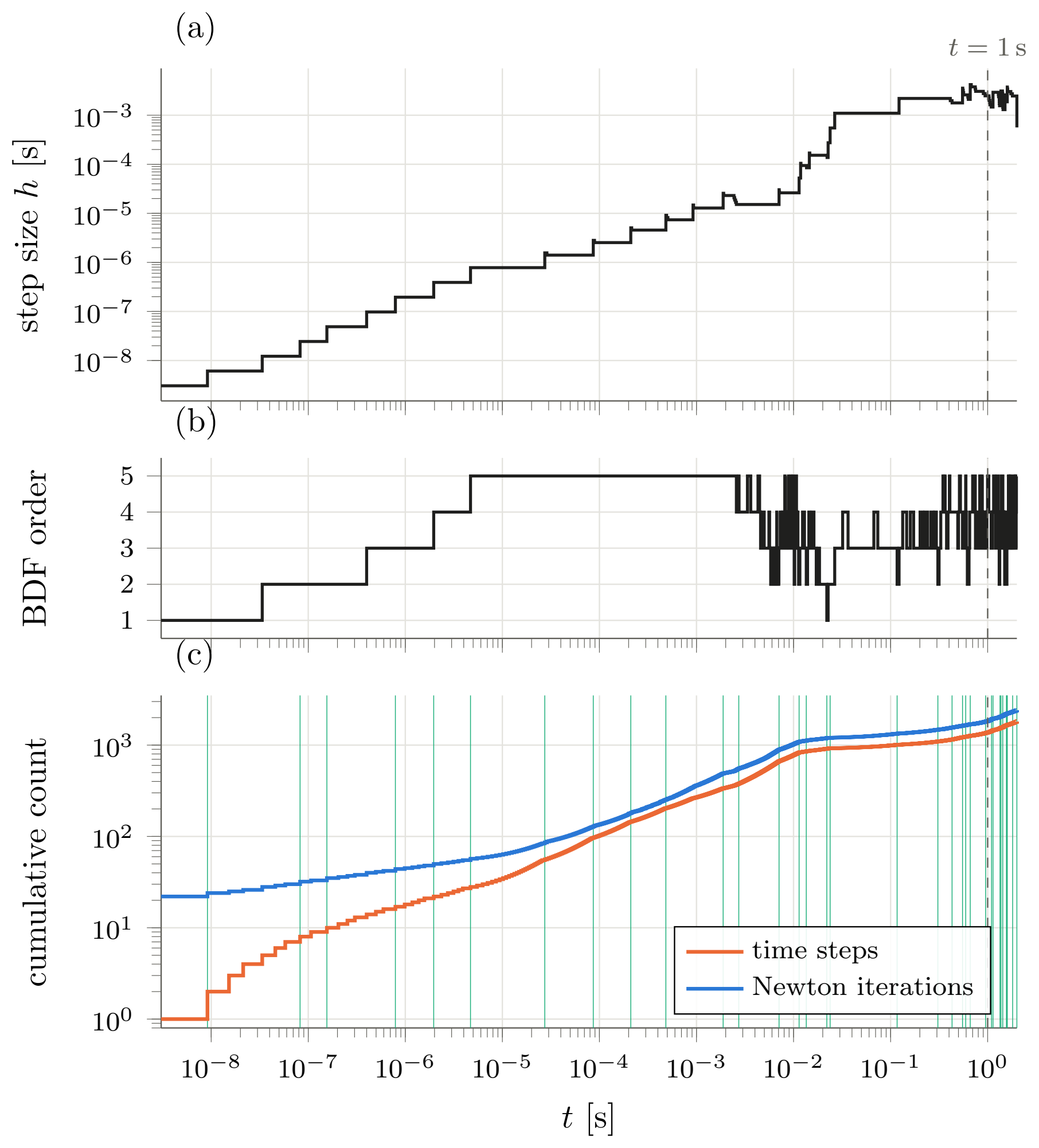

(a)
$t = 1\,\mathrm{s}$
step size h [s]
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
(b)
BDF order
5
4
3
2
1
(c)
cumulative count
10^{3}
10^{2}
10^{1}
10^{0}
time steps
Newton iterations
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
t [s]